



Từ tiến hóa di truyền *in vitro* đến kháng thuốc Flucytosine: Giải mã thách thức trong điều trị *Candida auris*

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Phuc-Loi Luu, PhD

Email: loilp@bvtvn.org.vn

Zalo: 0901802182



Viện Nghiên cứu Ứng dụng Khoa học Sức khỏe và Lão hóa – Bệnh viện Thống Nhất

Rapid *in vitro* evolution of flucytosine resistance in *Candida auris*

Trinh Phan-Canh ^{1,2}, Duc-Minh Nguyen-Le ³, Phuc-Loi Luu ^{3,4}, Narakorn Khunweeraphong^{1,2}, Karl Kuchler ^{1,2}

¹Max Perutz Labs, Vienna Biocenter Campus (VBC), Vienna, Austria

²Medical University of Vienna, Center for Medical Biochemistry, Vienna, Austria

³Institute for Applied Research in Health Sciences and Aging (ARiHA)-Thong Nhat Hospital, Ho Chi Minh City, Vietnam

⁴Mathematics Department, Faculty of Fundamental Sciences, University of Medicine and Pharmacy at Ho Chi Minh City, Ho Chi Minh City, Vietnam

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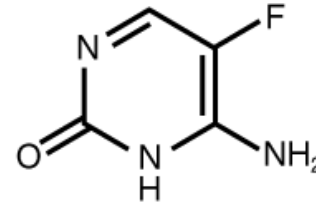
A JOURNAL BY THE AMERICAN SOCIETY FOR MICROBIOLOGY

Content

- Introduction to *Candida auris* and 5-fluorocytosine (5FC)
- RNA-seq and Whole genome Sequencing to profile gene expression and mutation
- *Invitro* evolution of genome experiment
- Why are R1, R2, R3, R4, R5 and R6 **resistant** to 5FC?
- Why is T1 **tolerant** of 5FC?
- Summary



https://vi.wikipedia.org/wiki/Candida_auris



Flucytosine
(5FC)

Candida auris: An Emerging Global Health Threat

- *Candida auris* is a **pan-antifungal-resistant fungal pathogen**.
- It has been responsible for **high-mortality** infection outbreaks, particularly in **hospital** and **healthcare** settings.
- While the prodrug 5-fluorocytosine (5FC) is considered one of the few treatment options, **resistance to it is pronounced**, limiting its use as a monotherapy.
- **The combination therapy involving 5FC shows promise** in treating these challenging infections.
- This pathogen can **rapidly develop resistance to antifungal drugs**, as demonstrated by the in vitro evolution of 5FC resistance in a clinical isolate after minimal drug exposure.
- **Understanding the mechanisms of resistance**, such as mutations in the *FUR1* and *FCY2* genes, is **crucial for developing effective treatment strategies**.



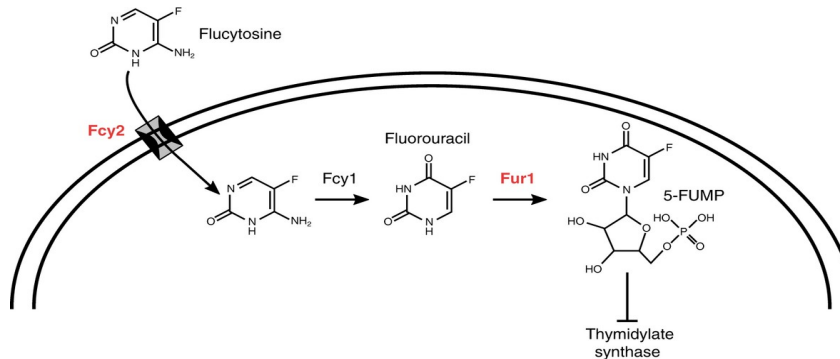
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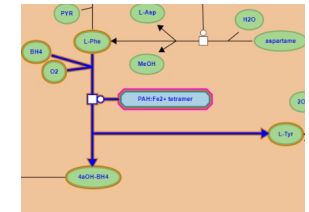
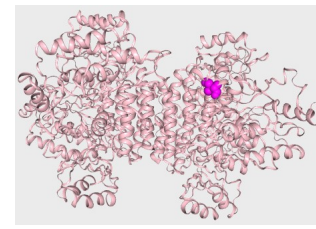
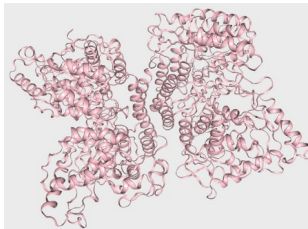
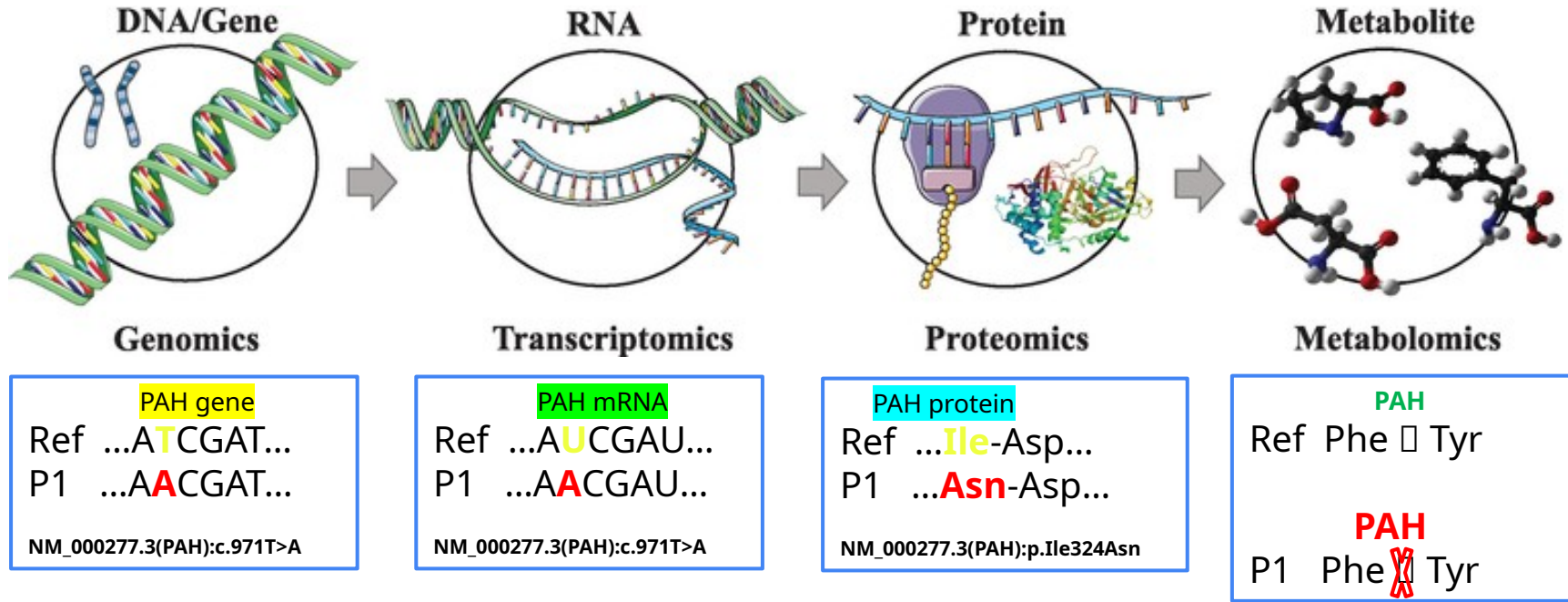
<https://www.scientificamerican.com>

5-Fluorocytosine (5FC): An Important Antifungal Drug

- 5-Fluorocytosine (5FC), or flucytosine, is an **antifungal prodrug**.
- 5FC is for **treating the pan-antifungal-resistant** pathogen *Candida auris*.
- 5FC is taken up by fungal cells and converted into the active antimetabolite **5-fluorouracil (5FU)**.
- 5FU is further converted into a toxic metabolite that **disrupts fungal RNA and DNA syntheses**.
- The clinical use of 5FC as an antifungal drug has been limited due to the **emergence of pronounced resistance**.
- The **development of resistance** to 5FC often involves mutations in genes like **FUR1** and **FCY2**.



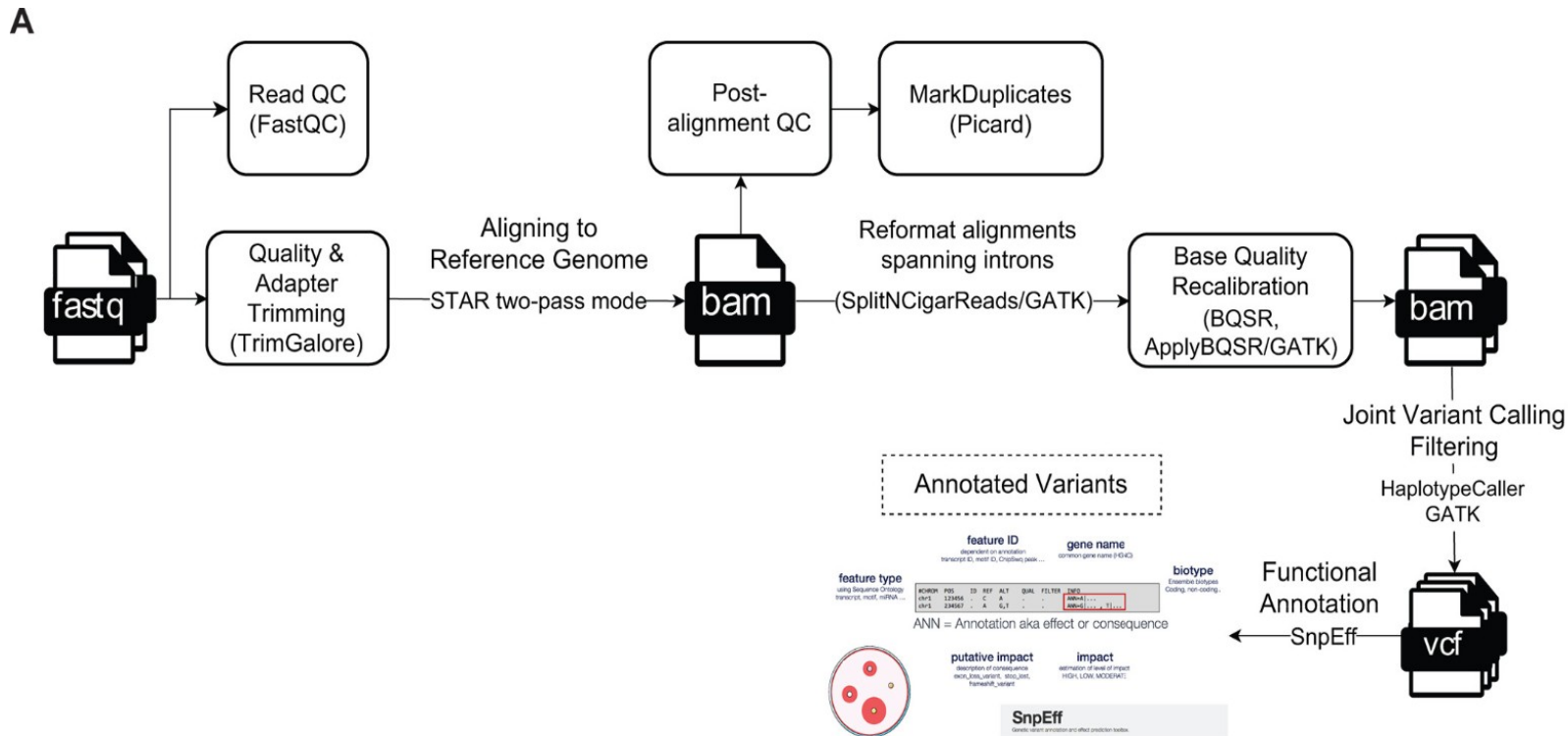
The link: Gene Mutation and Antifungal Drug Resistance



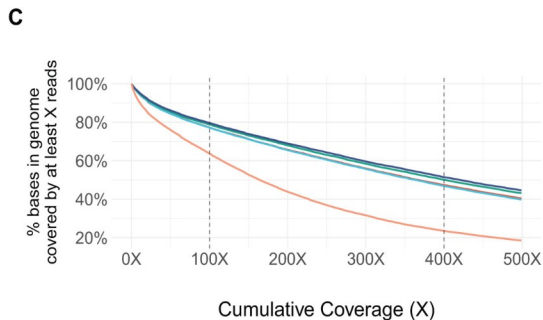
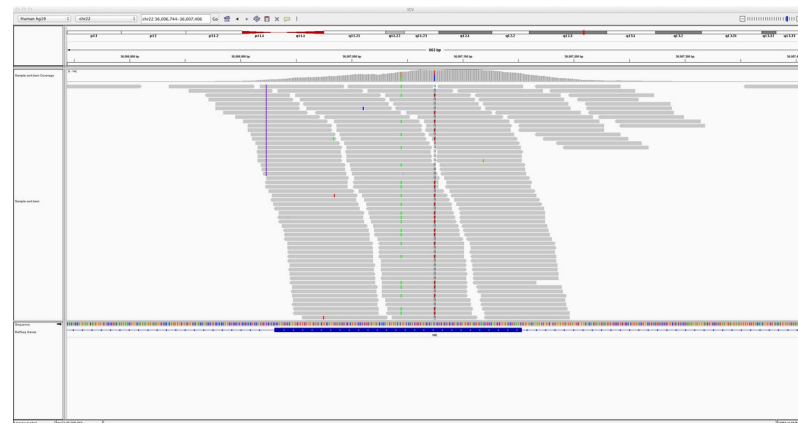
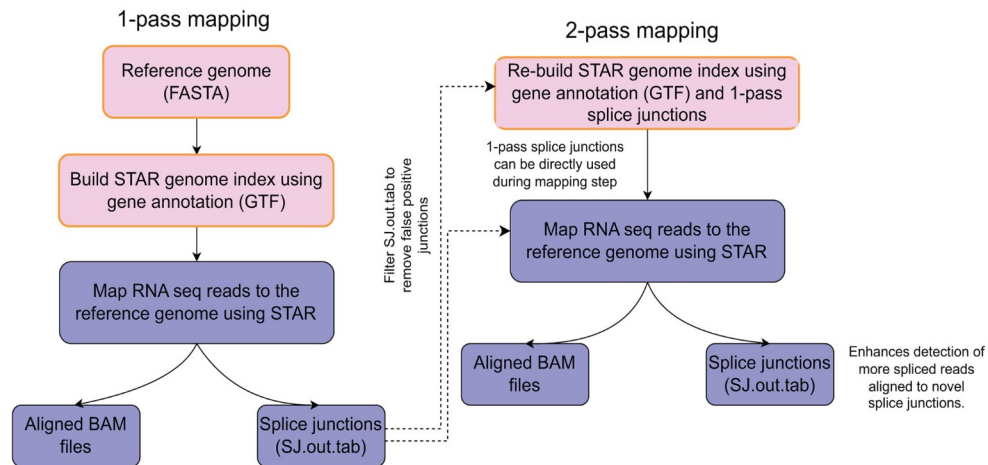
DNA sequencing for mutation identification

WT	GTAACCTTCCCAGAACCAGGAGAAGATGGAC	TG	GT	AA	GT	GA	TTACTGAAAATAACATGC
Cas9-B-2	GTACCCTTCCCAAAACCAGGAAAAAATGAAC	GGT	AA	TG	GATTACGGAAATTA	ACTGGC	
Cas9-B-8	GTACCCTTCCCAAAACCAGGAAAAAAGGGAC	GG	AA	TG	GATTACGGAAAATAC	CTGGC	
Cas9-B-20	GAACCCTTCCCAAAACCCAGAAAAAAAGGAAC	GG	AA	TG	GATTACGGAAATTAC	CTGGC	
Cas9-17-26	GAACCCTCCCCAAACCCAGAAAAAAAGGACT	GG	TA	AA	TG	ATACCGGAAATTAC	CTGGT
Cas9-17-27	GAACCCTTCCCAAAACCCAGAAAAAAAGGAAT	GG	AA	TG	AATACCGGAAATTAC	CTGGT	
Cas9-17-28	GAACCCTTCCCAAAACCCGGGAAAAAAAGGAAT	GG	TA	AA	TG	AATACCGGAAATTAC	CTGGT

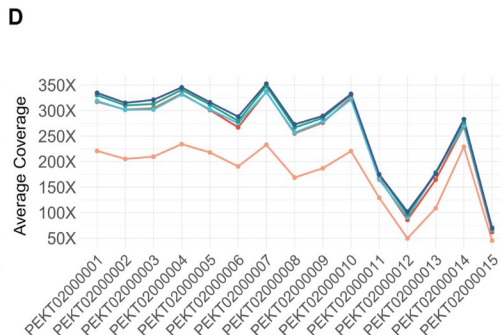
Computational workflow for SNP variant calling from RNA-seq data (1)



Computational workflow for SNP variant calling from RNA-seq data (2)

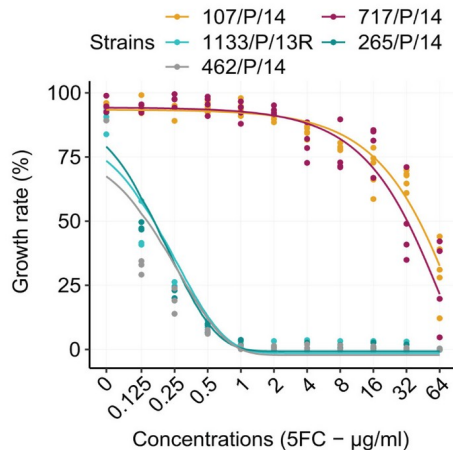


— WT — R1 — R2 — T1 — S1

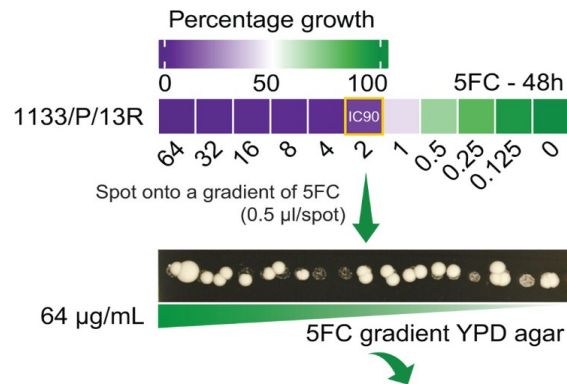


In vitro evolution of 5FC resistance in *C. auris*

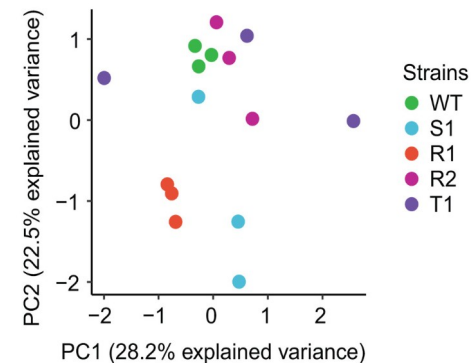
A



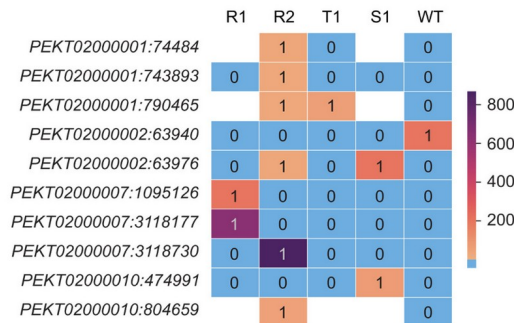
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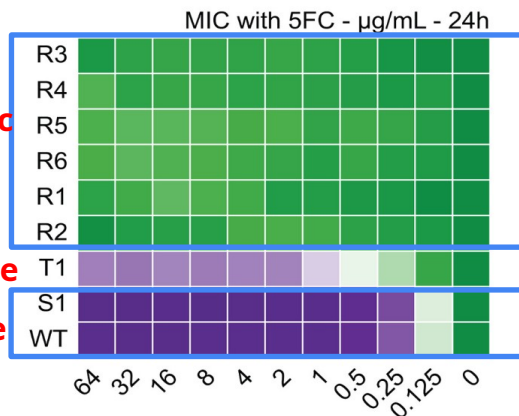
E



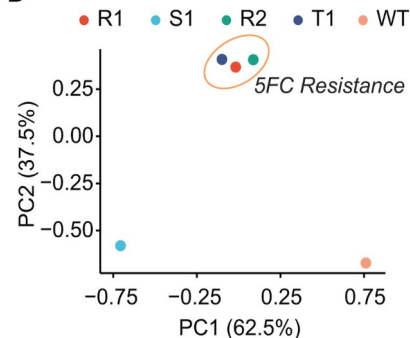
Resistance

Tolerance

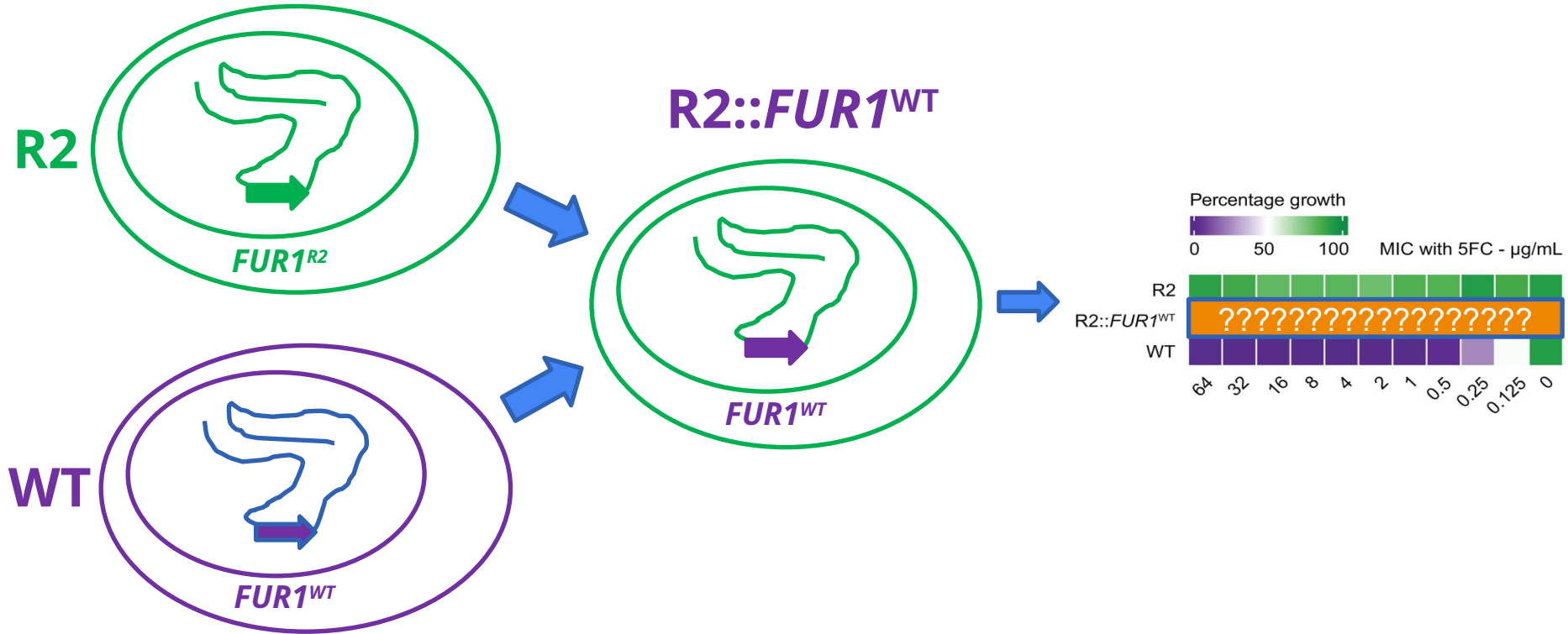
Sensitive



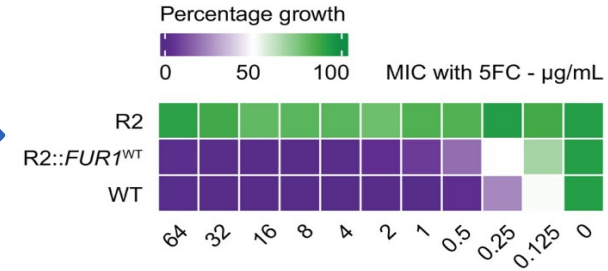
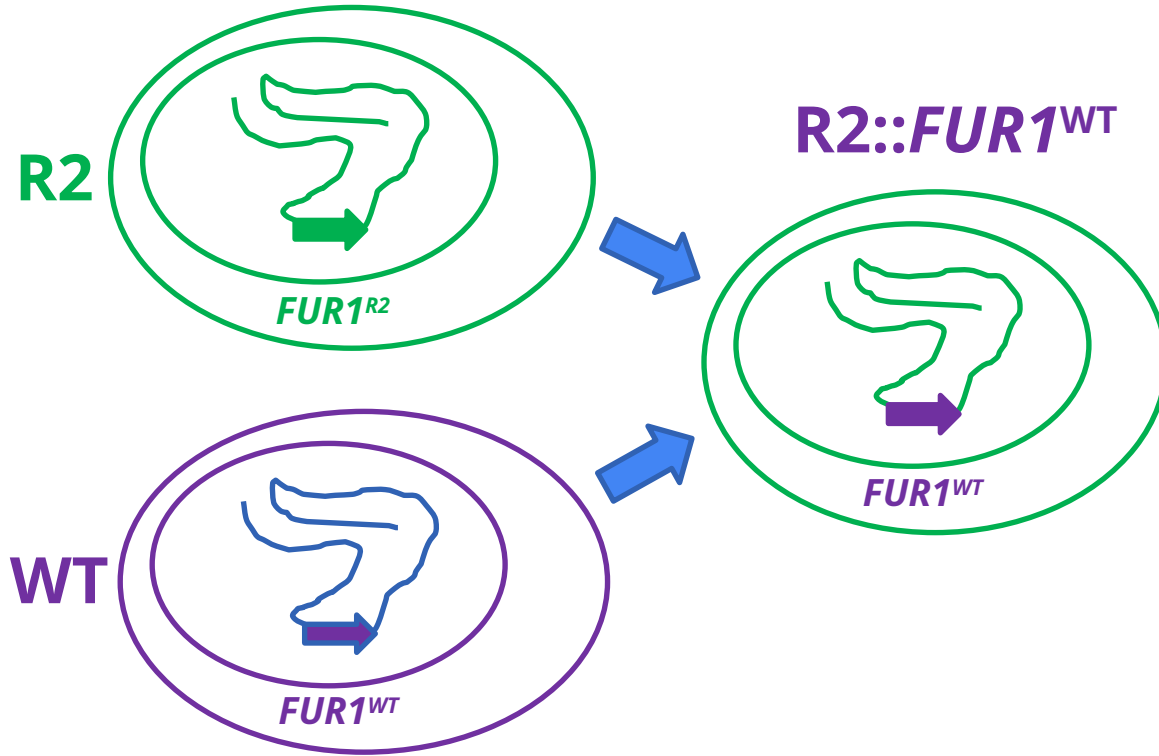
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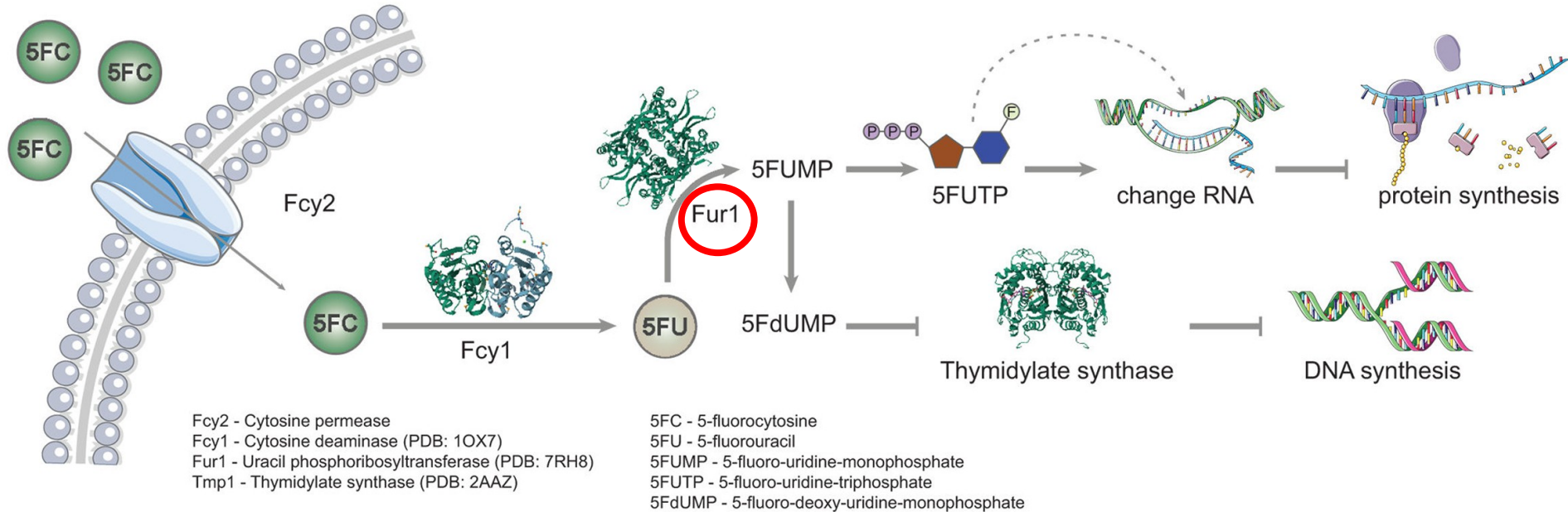
Creating R2::*FUR1*^{WT} using DNA recombinant technique



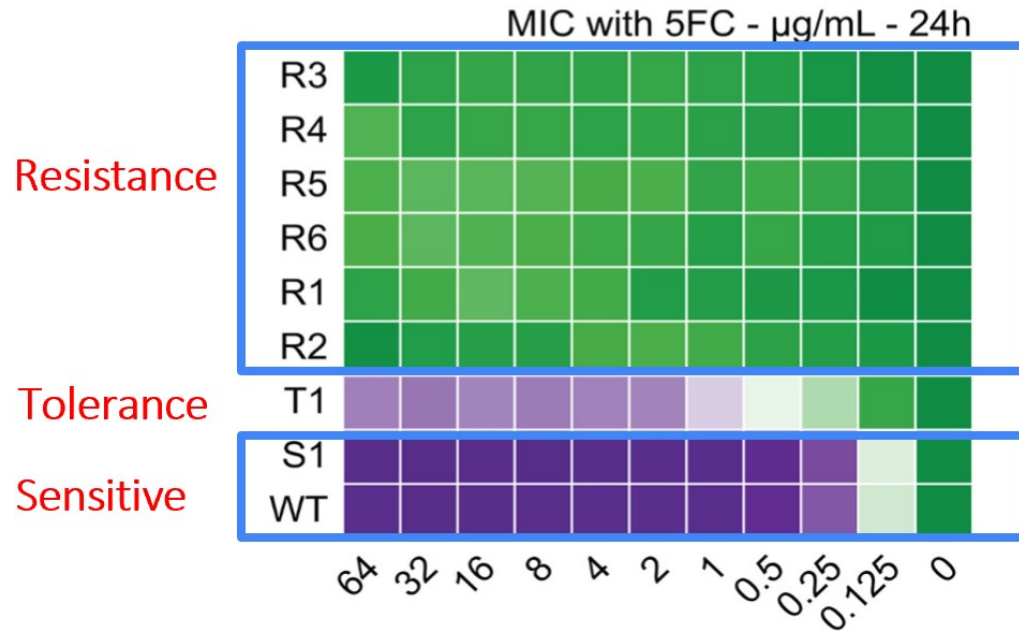
Creating R2::*FUR1*^{WT} using DNA recombinant technique



Rapid *in vitro* evolution of flucytosine resistance in *Candida auris*



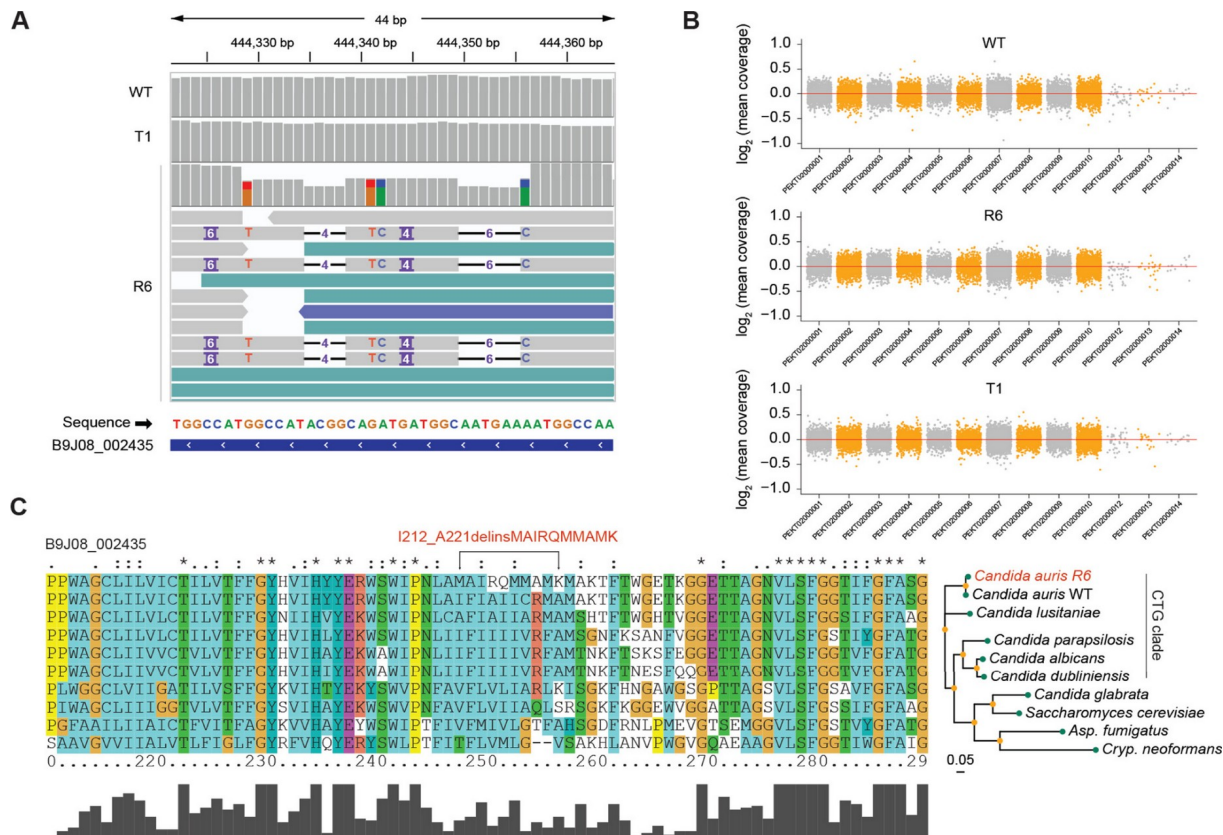
Why is R6 resistant to 5FC?



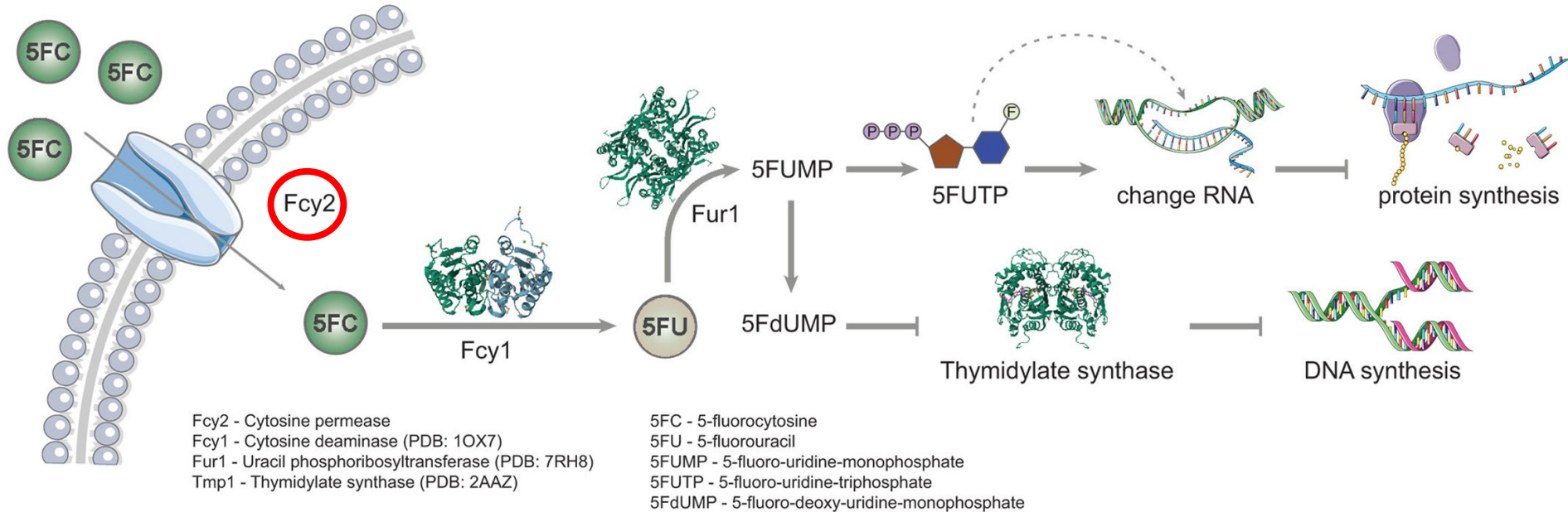
R6: Potential genetic variants corresponding to peptide changes in adapted clones compared to WT

Strain	Ref	Scaffold:Pos	GeneID	Gene	Effect	cDNA	aa	Method
R1	WT	PEKT02000007:1095126	B9J08_003129		Downstream	160G ^a > T		RNA-seq
R1	WT	PEKT02000007:3118177	B9J08_004076	<i>FUR1</i>	Missense	641G > C	R214T	RNA-seq, Sanger
R2	WT	PEKT02000002:63976	B9J08_000526		Missense	1795A > G	T599A	RNA-seq
R2	WT	PEKT02000010:804659	B9J08_005267	<i>RBF1</i>	Missense	81A > T	Q27H	RNA-seq
R2	WT	PEKT02000007:3118730	B9J08_004076	<i>FUR1</i>	Truncation	88C > T	Q30 ^a	RNA-seq, Sanger
R2	WT	PEKT02000001:74484	B9J08_000030	<i>RNT1</i>	Promoter	-439T > A		RNA-seq
R2	WT	PEKT02000001:743893	B9J08_000367	<i>ERG25</i>	Promoter	-483C > T		RNA-seq
R2 ^a	WT	PEKT02000001:790465	B9J08_000387 ; B9J08_000388	tRNA; <i>NDT80</i>	Intergenic	-1368T > A		RNA-seq
T1 ^a	WT	PEKT02000001:790465	B9J08_000387 ; B9J08_000388	tRNA; <i>NDT80</i>	Intergenic	-1368T > A		RNA-seq
S1	WT	PEKT02000010:474991	B9J08_005113	<i>TFB2</i>	Promoter	-387A > T		RNA-seq
S1	WT	PEKT02000002:63976	B9J08_000526		Missense	1795A > G	T599A	RNA-seq
R6	WT	PEKT02000006:444342	B9J08_002435	<i>FCY2</i>	Missense	650T > G	I217S	WGS ^c
R6	WT	PEKT02000006:444356	B9J08_002435	<i>FCY2</i>	Missense	636T > G	I212M	WGS
R6 ^b	WT	PEKT02000008:8518	B9J08_004105		Missense	482A > T	D161V	WGS
R6 ^b	WT	PEKT02000008:8523	B9J08_004105		Missense	487A > G	N163D	WGS
T1	WT	PEKT02000008:8654	B9J08_004104 - B9J08_004105		Intergenic	-2887A > T		WGS
T1	WT	PEKT02000008:8662	B9J08_004104 - B9J08_004105		Intergenic	-2895T > C		WGS
T1	WT	PEKT02000004:842355	B9J08_001924	<i>SEC27</i>	Upstream	-2996A > G		WGS
R6	WT	PEKT02000006:444356	B9J08_002435	<i>FCY2</i>	Missense	636_663delins-5'-GGCCATAC GGCAGATGATG GCAATGAAA-3'	I212_A221delins MAIRQMMAMK	Sanger
R3	WT	PEKT02000007:3118730	B9J08_004076	<i>FUR1</i>	Truncation	88C > T	Q30 ^a	Sanger
R4	WT	PEKT02000007:3118685	B9J08_004076	<i>FUR1</i>	Frameshift	Δ4nt175fs		Sanger
R5	WT	PEKT02000007:3118730	B9J08_004076	<i>FUR1</i>	Truncation	88C > T	Q30 ^a	Sanger

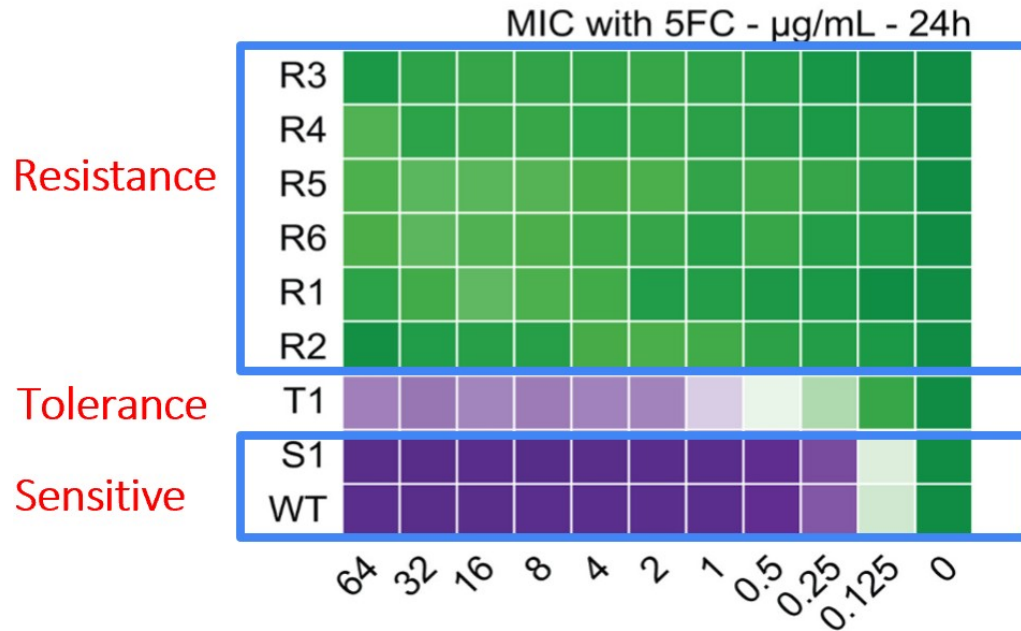
R6: A mutation in the FCY2.2 homolog (B9J08_002435) contributes to 5FC resistance



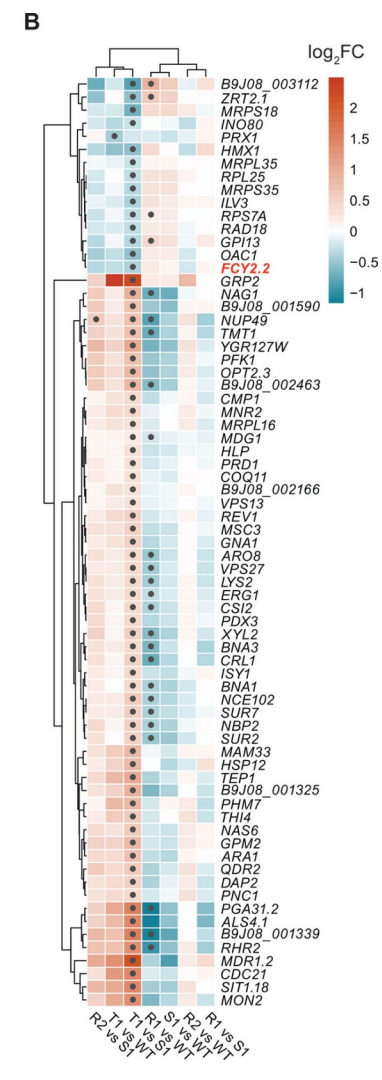
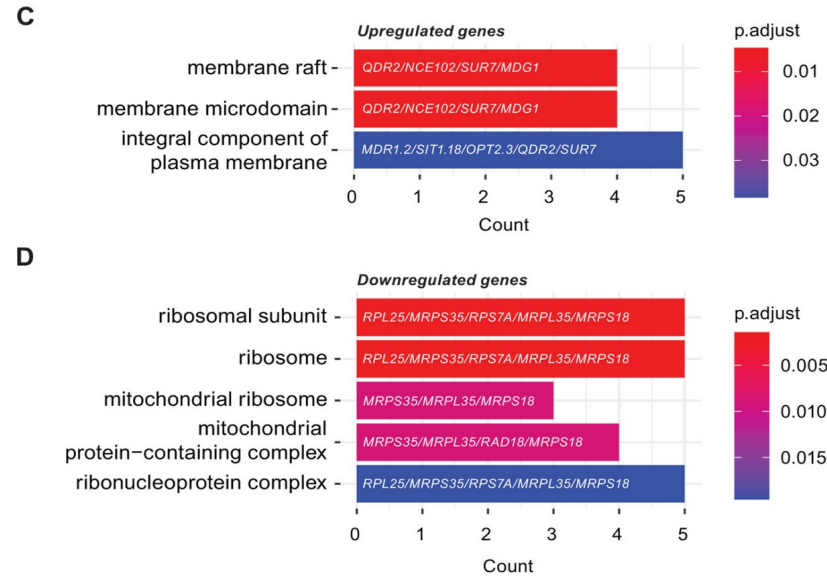
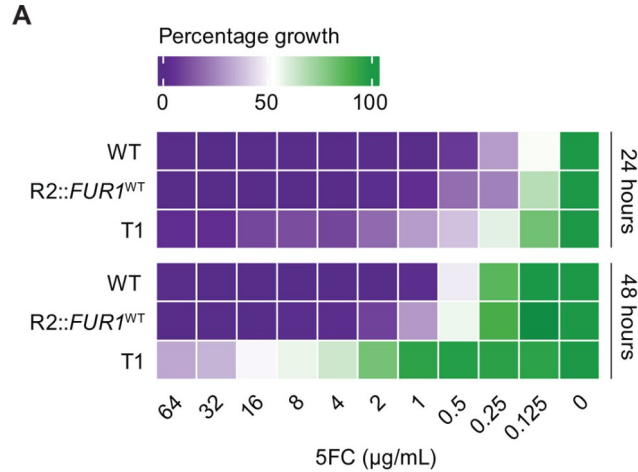
Rapid *in vitro* evolution of flucytosine resistance in *Candida auris*



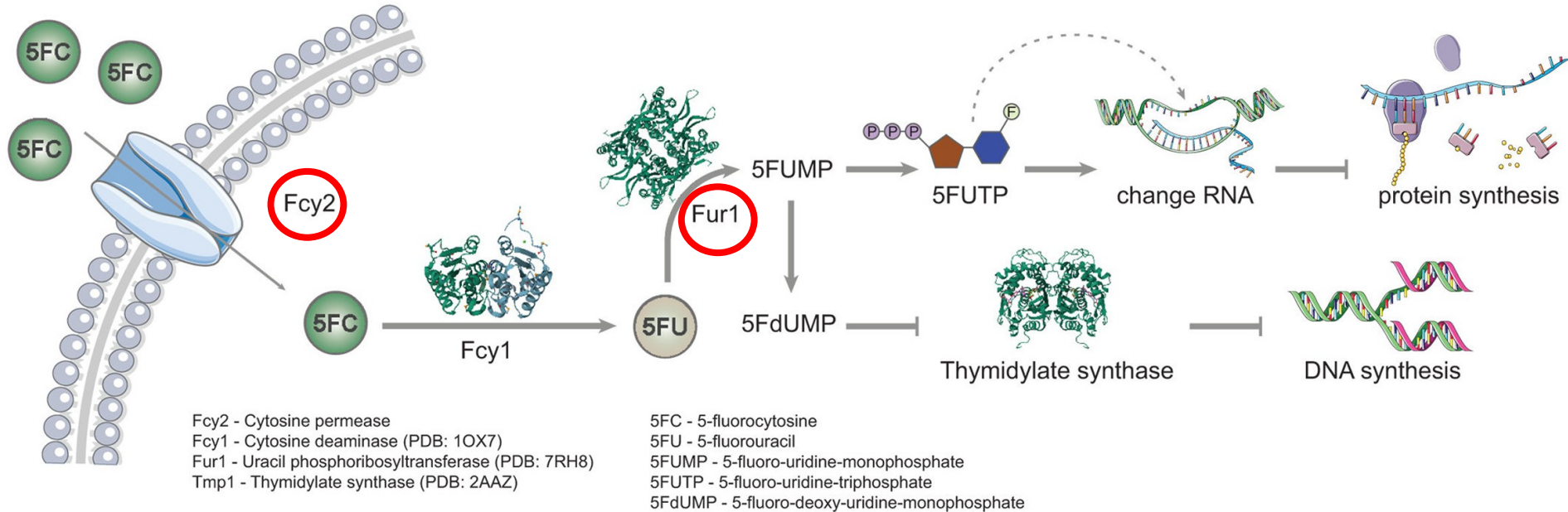
Why is T1 **tolerant** of 5FC?



The 5FC tolerance is associated with gene expression profiles



Rapid *in vitro* evolution of flucytosine resistance in *Candida auris*



Keynotes

Problem: Treating *Candida auris* infections with 5-fluorocytosine (5FC) alone is problematic due to the rapid development of resistance.

Recommendation: Combination therapy including 5FC is essential for effective treatment.

Need: Effective tools are required to predict the potential for 5FC resistance development when using combination therapy.

Proposed Solution: A bioinformatic workflow using RNA-seq data can monitor both gene expression and genetic changes simultaneously in clinical isolates.

Benefit: This workflow aids in tracking and potentially anticipating the emergence of clinical 5FC resistance.



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Phuc-Loi Luu, PhD



Nguyen Le Duc Minh, Dr.
Bioinformatic Specialist



Ngo Dai Phu, Msc.
Project Manager



Dao Khuong Duy, Msc.
Bioinformatic Specialist



Hong Ong, Msc.



Thanh Nguyen, Bsc.
Bioinformatic Specialist



Giau Le, Bsc.
Software Engineer



Ho Phu Quy, Bsc.
Bioinformatic Specialist



Pham Hieu Dan, Bsc.
Bioinformatic Specialist



Nguyen Manh Hung, Bsc.
Bioinformatic Specialist



Vuong Vu Hoang, Bsc.
Bioinformatic Trainee



Nguyen Quang Khai, Bsc.
Bioinformatic Specialist



Kim Van, Bsc.
Bioinformatic Specialist



Huy Ha, Bsc.
Bioinformatic Specialist



Nguyen Thanh Cong, Msc.
Bioinformatic Trainee

Thank you for your listening
Q & A

Phuc-Loi Luu, PhD

Email: loilp@bvtv.org.vn

Phone/Zalo: 0901802182